



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Fall 2023

Paper: Information Security

Course Code: CC-312

(CLASH)

Roll No. 063067

Time: 3 Hrs.

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

1. Encrypt the plaintext "EXAM" using Hill Cipher, where $k = (11 \ 8 \ 3 \ 7)$
2. Show that DES decryption is, in fact, the inverse of DES encryption.
3. What characteristics are needed in a secure hash function?
4. Using keyword "ANTICIPATION", Plaintext "You Should Defend the East Wall of the Castle". You should find Ciphertext using Autokey Cipher.
5. What is the difference between a monoalphabetic cipher and a polyalphabetic cipher? Also Explain Playfair Cipher.
6. What is the purpose of the S-boxes In DES? Explain

Q.2. Answer the following questions.

(3x10=30)

1. In a public-key system using RSA, you intercept the ciphertext $C = 10$ sent to a user whose public key is $e = 5$, $n = 35$. What is the plaintext M ? Determine p , q such that $n = p \cdot q$ and $M = C^d \bmod n$.
2. Briefly explain following public key distribution techniques with diagram
 - a) Public Key Directory
 - b) Public Key Authority
3. Decrypt the Ciphertext $C = (DC6A)_{\text{HEX}}$ using DES Algorithm. The description of this algorithm is given as:

$C = (1101 \ 1100 \ 0110 \ 1010)_2$ and $K = (1010 \ 1011 \ 1100 \ 1101)_2$. We have customized the standard DES Algorithm with the following assumptions (denoted by A)

A#1: There is only ONE round for encryption and decryption.

A#2: Plaintext length = key length = 16 bits. ($L_0 = R_0 = 8$ bits)

A#3: The Sub-key K_1 is of length 12 bits.

A#4: E-Box takes 8 bits data as input and generates 12 bits data as output (see Table # 3)

A#5: There are two Compression boxes S_1 and S_2 both takes 6-bit as input and generates 4-bit output

(see Table # 4 and Table # 5)

A#6: P-Box takes 8-bit data as input and generates 8-bit data as output. (see Table # 6)

A#7: Key generator module operation are as follows:

A#7.1: PC1 box takes 16-bit key as input and generates 14-bit key as output by two dropping parity bits.

(Every 8th bit is parity bit: see Table # 7)

A#7.2: Each Shift box (C and D) is of size 7-bit and contents of C = { C₁ } and D = { D₁ } are circularly shifted to left by one bit.

A#7.3: PC2 box takes 14-bit key as input and generates 12-bit sub-key as output by dropping 3rd and 13th bits.

All referenced tables are given below.

Table # 1 : Initial Permutation

Actual bit location	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Moves to location	4	9	12	14	10	2	8	11	6	1	16	3	15	5	13	7

Table # 2 : Inverse of Initial Permutation

Actual bit location	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Moves to location	10	6	12	1	14	9	16	7	2	5	8	3	15	4	13	11

Table # 3 : E-Box (Expansion and Permutation Box)

Actual bit location	7	5	2	4	2	8	1	4	5	3	6	7
Moves to location	1	2	3	4	5	6	7	8	9	10	11	12

Table # 4 : S1 Compression Box

00	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
01	0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
10	4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
11	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

Table # 5 : S2 Compression Box

00	15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
01	3	13	4	7	15	2	8	14	12	0	1	10	6	0	11	5
10	0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
11	13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9

Table # 6 : P-Box (Last step of 'P' function)

Actual bit location	3	6	2	7	1	4	8	5
Moves to location	1	2	3	4	5	6	7	8

Table # 7 : PC1-Box (Key generator module) maps 16 bits on 14 bits

Actual bit location	10	6	12	1	14	9	7	2	5	3	15	4	13	11
Moves to location	1	2	3	4	5	6	7	8	9	10	11	12	13	14